**MATH 1271** **Name:** _____
**Fall 2026**
**Quiz 8 - 10/08 Content**
**10/13/2026**
**Time Limit: 5 minutes** **Student ID:** _____

This quiz consists of 1 questions across 2 pages worth 5 points.
**Rules:**

- You have the indicated amount of time above to complete the quiz and you must not begin working on the quiz until instructed to do so.
- You must show all of your work to receive full credit, unless stated otherwise. **On problems that require work to be shown, answers that are mysterious or unsupported will not receive full credit.**
- You may not use a calculator, textbook, notes or any other external resources.
- Please write your initials at the top of every page in the event the pages become separated.
- If you run out of space on a page, feel free to use the back of the page to continue your work.
- You understand that if you violate any of these or the University's rules, you will receive a score of zero on this quiz and you may be subject to further disciplinary action.

Following the University of Minnesota's policy on academic honesty, I pledge that the work shown on this quiz is entirely my own and that I have followed the rules stated above.

Signature: _____

| Question | Points |
|---|---|
| 1 | 5 |
| Total: | 5 |

1. A certain function $y = p(x)$ has its local linearization at $a = 3$ given by $L(x) = -2x + 5$.

   (a) (1 point) What are the values of $p(3)$ and $p'(3)$? Why?

   **Solution:** $p(3) = L(3) = -2 \cdot 3 + 5 = -6 + 5 = -1$ and $p'(3) = L'(3) = -2$ since the linearization of a function at a point agrees with the function (they meet) and has the same derivative (is tangent).

   (b) (2 points) Estimate the value of $p(2.79)$.

   **Solution:**

$$\begin{aligned} p(2.79) &\approx L(2.79) = -2 \cdot 2.79 + 5 && (1)\\ &= -5.58 + 5 && (2)\\ &= -0.58 && (3) \end{aligned}$$

   (c) (2 points) Suppose that $p''(3) = 0$ and you know that $p''(x) < 0$ for $x < 3$. Is your estimate in (b) too large or too small?

   **Solution:** Since the second derivative is negative for points less that 3, we know that the graph of the function falls below the tangent lines near and to the left of $x = 3$, so our estimate is too large.